Lecture 15 — 02.12.2025

Theorem (variational and minimization problems) Let $a: V \times V \to \mathbb{R}$ be a symmetric and positive definite bilinear form of the variational problem

$$\text{Find } u \in V \text{ s.t. } a(u,v) = F(v) \text{ for all } v \in V. \qquad (VP)$$

Define $J: V \to \mathbb{R}$ by $J(w) = \frac{1}{2} a(w,w) - F(w)$. Then the minimization problem

$$\text{Find } u \in V \text{ s.t. } J(u) \leq J(v) \text{ for all } v \in V \qquad (MP)$$

has the same solution (if it exists) as (VP).

Proof (VP) $\Rightarrow$ (MP)

Let u solve (VP) and let $v \in V$ be arbitrary. Define $w := v - u \in V$.

$$\Rightarrow J(v) = J(u+w) = \frac{1}{2} a(u+w, u+w) - F(u+w)$$

$$\overset{a \text{ is bilinear}}{=} \frac{1}{2}\left[a(u,u) + a(w,u) + a(u,w) + a(w,w)\right] - F(u) - F(w)$$

a is symmetric

$$= \underbrace{\frac{1}{2} a(u,u) - F(u)}_{J(u)} + \underbrace{a(u,w) - F(w)}_{=0 \ (VP)} + \frac{1}{2} \underbrace{a(w,w)}_{\geq 0} \geq J(u)$$

a is pos. def.

(MP) $\Rightarrow$ (VP)

Assume that u solves (MP). Consider an arbitrary $v_0 \in V$ and $\alpha \in \mathbb{R}$.

We define $j: \mathbb{R} \to \mathbb{R}$, $j(\alpha) := J(u + \alpha v_0)$

$\Rightarrow \quad j(0) = J(u) \leq J(u + \alpha v_0) = j(\alpha) \quad \forall \alpha \in \mathbb{R}$

$\Rightarrow$ j has a global minimum in $\alpha = 0$, hence $j'(0) = 0$

Note: $j(\alpha) = J(u + \alpha v_0) = \frac{1}{2} a(u,u) - F(u) + \alpha a(u, v_0) - \alpha F(v_0) + \frac{1}{2} \alpha^2 a(v_0, v_0)$

$\Rightarrow \quad j'(\alpha) = a(u, v_0) - F(v_0) + \alpha a(v_0, v_0)$

$\Rightarrow \quad 0 \overset{!}{=} j'(0) = a(u, v_0) - F(v_0)$ for arbitrary $v_0 \in V \Rightarrow$ (VP)

Example Consider $-u''(x) = f(x)$ in $(0,1)$, $u(0) = 0 = u(1)$

$\Rightarrow$ (bi)linear form $a(u,v) = \int_0^1 u'(x) v'(x)\, dx$, $F(v) = \int_0^1 f(x) v(x)\, dx$

Let us choose the finite-dimensional subspace

$$V_n = \text{span}\{\underbrace{\sin(j\pi x)}_{\varphi_j(x)} \mid j = 1, \dots, n\} = \{w: (0,1) \to \mathbb{R} \mid w(x) = \sum_{j=1}^{n} \alpha_j \sin(j\pi x), \alpha_j \in \mathbb{R}\}$$

Note φ_j is once weakly (even classically) differentiable and satisfies $\varphi_j(0) = 0 = \varphi_j(1)$

$$\Rightarrow \quad a(\varphi_i, \varphi_j) = \int_0^1 \underbrace{(\pi i \cos(i\pi x))}_{\varphi_i'(x)} \cdot \underbrace{(\pi j \cos(j\pi x))}_{\varphi_j'(x)} \, dx = \ldots = \begin{cases} \frac{j^2}{2}\pi^2 & , i = j \\ 0 & , \text{else} \end{cases}$$

$\Rightarrow$ we therefore obtain a diagonal matrix A_n (in $A_n \alpha = f_n$)

For the linear form F we obtain $F(\varphi_j) = \int_0^1 f(x) \sin(j\pi x)\, dx = (f_n)_j$

$\Rightarrow$ solving $A_n \alpha = f_n$ yields $\alpha_j = \frac{2(f_n)_j}{j^2\pi^2}$ and $u_n(x) = \sum_{j=1}^{n} \alpha_j \sin(j\pi x)$

Here, the choice of the subspace V_n was (computationally) convenient, since it resulted in a diagonal A_n! (Reason: orthogonal basis $\langle \varphi_i, \varphi_j \rangle_{L^2(0,1)} = \delta_{ij}$)

Unclear, if the choice of V_n leads to an accurate approximation.

<u>Summary and conclusion:</u>

1) If a particular Galerkin method is "good" depends strongly on the choice of the subspaces $(V_n)_{n\in\mathbb{N}}$, $V_n \subset V$. More specifically, we require

$$(*) \qquad \inf_{v_n \in V_n} \| u - v_n \|_V$$

$\hookrightarrow$ not necessary, but we focus on this case

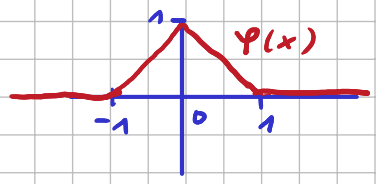
to be "small".

2) In addition (potentially in conflict) to 1), we are interested in finding subspaces $(V_n)_{n\in\mathbb{N}}$ such that the corresponding linear system $A_n \alpha = f_n$ is "easy" to solve (e.g., we want A_n to be sparse) and A_n is "easy" to set up / "assemble"

3) As in the FDM, we are interested in the behavior of v_n in $(*)$ if $n \to \infty$ (convergence?)

$\rightsquigarrow$ one choice that complies with these requirements are hat functions:

$$\varphi(x) = \begin{cases} x+1 & \text{if } x \in [-1,0] \\ 1-x & \text{if } x \in (0,1] \\ 0 & \text{else} \end{cases}$$



Finite elements

<u>Strategy</u>: 1) Decompose $\overline{\Omega}$ into finitely many parts $\overline{\Omega} = \bigcup_{i=1}^{n_e} \overline{\Omega_i}$ in an admissible way

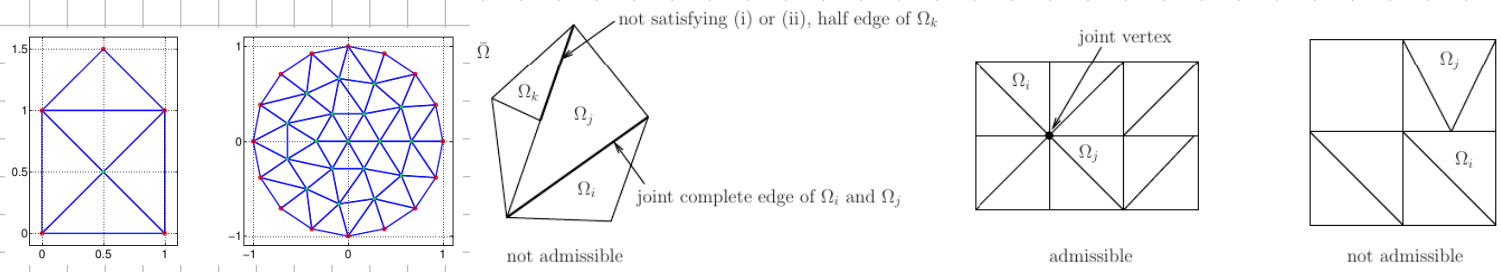
$\rightsquigarrow$ we refer to $\overline{\Omega_i}$ as an element in this decomposition of $\overline{\Omega}$

2) On each element Ω_i we focus on a certain (finite dimensional) space of functions supported on the element (polynomials up to a certain degree)

$\Rightarrow$ together we refer to 1) and 2) as finite element

Definition (Admissible decomposition) Assume a given domain $\Omega \subset \mathbb{R}^d$ with polygonal boundary and let $\overline{\Omega} = \bigcup_{i=1}^{n_e} \overline{\Omega_i}$ be a decomposition of $\overline{\Omega}$ into elements $\{\overline{\Omega_i}\}_{i=1,\dots,n}$.

- $d=1$: If $\Omega = (a,b) \subset \mathbb{R}$ and $a = x_0 < x_1 < \dots < x_N < x_{N+1} = b$, then an <u>admissible decomposition</u> is given by a union of non-overlapping but adjacent intervals $\Omega_i = (x_{i-1}, x_i)$, $i = 1, \dots, n_e = N+1$.
- $d=2$: A decomposition of Ω into triangles (or convex quadrilaterals, ...) $\{\Omega_i\}_{i=1}^{n_e}$ is <u>admissible</u> if for each i,j <u>exactly one</u> of the following cases is true:
  - i) $\Omega_i = \Omega_j$
  - ii) $\overline{\Omega_i} \cap \overline{\Omega_j}$ is a joint complete edge of both Ω_i and Ω_j
  - iii) $\overline{\Omega_i} \cap \overline{\Omega_j}$ is a joint vertex of Ω_i and Ω_j
  - iv) $\overline{\Omega_i} \cap \overline{\Omega_j} = \emptyset$



- $d=3$: A decomposition into tetrahedrons (or hexahedrons, pyramids, ...) $\{\Omega_i\}_{i=1}^{n_e}$ is <u>admissible</u> if for each i,j <u>exactly one</u> of the following cases is true:
  - i) $\Omega_i = \Omega_j$
  - ii) $\overline{\Omega_i} = \overline{\Omega_j}$ is a joint complete face of both Ω_i and Ω_j (triangle, quadrilateral, ...)
  - iii) $\overline{\Omega_i} \cap \overline{\Omega_j}$ is a joint complete edge of both Ω_i and Ω_j
  - iv) $\overline{\Omega_i} \cap \overline{\Omega_j}$ is a joint vertex of Ω_i and Ω_j
  - v) $\overline{\Omega_i} \cap \overline{\Omega_j} = \emptyset$